Homework 6: Policy Gradient Reinforcement Learning

CS 1470

Due November 22nd, 2019 at 11:59pm

1 Conceptual Questions

1. What are some of the differences between the REINFORCE algorithm (Monte-Carlo method) and the Advantage Actor Critic? (3-5 sentences)

   Hint: Remember to discuss the differences in the loss functions between the two methods
   Hint: Discuss the variance differences of the gradient estimates

   Answer: The main differences come from loss functions and variance. For reinforcement learning, the loss is defined as $L(s) = -\Sigma_{t=1}^{T} \nabla \log p(a_t|s_t) D(s_t, a_t)$ where $D(s_t, a_t)$ has a high variance since it depends on the subsequent actions. For advantage actor critic learning, the D term is replaced by $D(s_t, a_t) - V(s_t)$, where $V(s_t)$ can be approximated by a value network. The total loss can be written as $L_total = L_{actor} + L_{critic}$, where $L_{actor} = \log p(a_t|s_t)(D(s_t, a_t) - V(s_t))$ and $Lcritic = (D(s_t, a_t) - V(s_t))^2$. This modification of loss guarantee a smaller variance for the loss of actor function and imporve the performance of actor-critic learning.

2. In reinforcement learning, we often discuss the exploration-exploitation tradeoff. We must balance acquiring new knowledge about our environment (exploration) while simultaneously trying to maximize our return (utilizing all the knowledge we've collected thus far). How do policy gradient methods automatically perform this tradeoff? (2-4 sentences)

   Answer: The policy gradient method collects the training episodes from random sampling which ensures the exploration its environment. And the same time, it learns from these collected episodes by learning a better policy function from them which maximize the return from these data. Intuitively, the loss function function could guarantee the future is more predictable while the reward is maximizing.

3. What advantages do policy gradient methods have over deep Q learning methods? What advantages do deep Q learning methods have over policy gradient methods? (3-5 sentences)

Answer: The advantages of Reinforce over DQN are:

policy function is generally easier to learn than the Q function since the space could be really large for most of the real world circumstances.

There is a tradeoff between exploration and exploitation

The policy is learned from stochastic behavirors.

It converges faster than DQN

But the DQN has its unique advantages over Reinforcement learning:

It is not that easily be trapped by local minimum.

It is more stable in terms of training.

DQN could simultaneously be updated after every step, which is good for faster convergence and batched training.

2 Ethical Implications

1. One of the most important and quickly-developing uses of reinforcement learning is for self-driving vehicles. These autonomous vehicles stand to revolutionize industries like ride-sharing, trucking, and package delivery, potentially automating over 4 million currently existing jobs. Check out some news reports on this issue, ranging from the pessimistic to the neutral to the optimistic (pp. 10-12).

   Do the potential economic benefits of autonomous vehicles outweigh the labor costs? Why or why not? (2-4 sentences)

   Answer: In the long term, the potential economic benefits of autonomous vehicles would outweigh the labor costs. Take airplane for example, all of modern airplanes are using self-control system running by computer, and it turns out the airplane is the safest transportation way. Similarly, the self-driving system could definitely protect people from accidents by reducing human errors. Technically, the shortage of self-driving cars comes from the uncertainty of the algorithm, low interpretablility of machine acting, but these shortages will eventually be solved and conquers by scientists. Once we conquer these problems, this technology would greatly reduce the accident rate, thus it will save a lot of expenditure in medical care, insurance, or other related area. By that time, labor costs will also be greatly reduced for maintaining self-driving cars.

2. Within the realm of speculative fiction, much has been written about the obligations of scientists to the sentient beings that they create (Mary Shelley's *Frankenstein*, Ted Chiang's *Lifecycle of Software Objects*, and **many** others). Some computer scientists and philosophers have argued that with recent advances, RL algorithms represent a kind of sentience, and that humans should take the suffering of RL algorithms into account. This has even led to the creation of groups like People for the Ethical Treatment of Reinforcement Learners.

Do you agree or disagree with this viewpoint? Do we as humans currently have moral obligations to treat RL algorithms ethically? Why or why not? If not, what would change your mind? (3-5 sentences)

Answer: I agree with this point of view. Any advancement in science and technology must be oversaw and regulated by ethics, this is especially correct in medicine. RL algorithm is not frighting, but the way to apply the RL algorithms must be oversaw, since it could direct machine to achieve something that machines could never achieve before. Things like self-driving or robotics are greatly changing the world we live in, so to make sure that bright impacts of these technologies could bring outweigh the drawbacks, we humans must have moral obligations to trest RL ethically.